

Day 1

Info - Hypervisor Overview

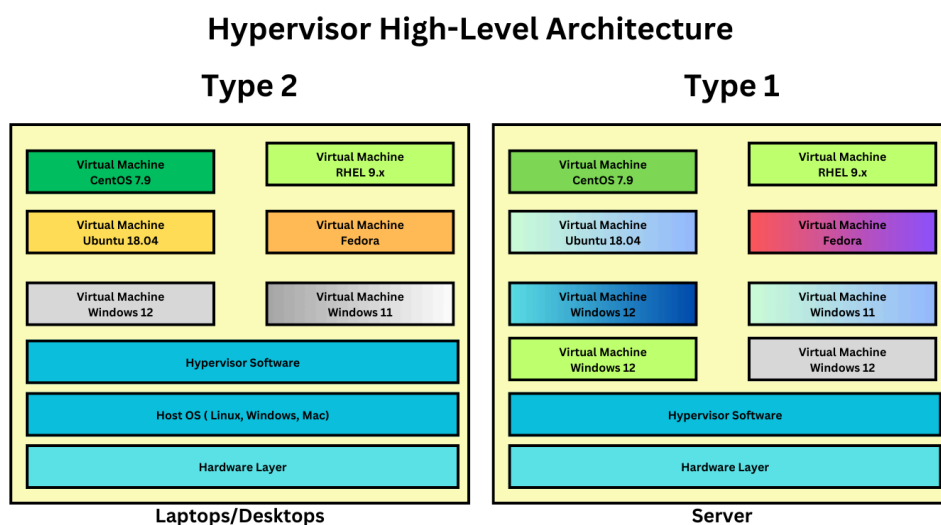
- nothing but virtualization
- with virtualization/hypervisor software, we can run multiple OS side by side on the same laptop/desktop/workstation/server
- there are 2 types of Hypervisors
 1. Type 1
 - a.k.a Baremetal Hypervisor
 - is used in Workstation & Server
 - performance wise it offers almost near native performance (3% less performance as opposed to running a OS on direct H/W)
 - this can be installed directly on top of a H/W with no Operating System
 - examples
 1. VMWare vSphere(vCenter) - Paid
 2. Linux KVM (opensource & Free)
 3. Microsoft Hyper-V
 2. Type 2
 - a.k.a Hosted Hypervisor
 - this can be installed only top of a Host Operating System (Windows, Mac OSX, Linux, etc)
 - is used in Workstation, Desktops & Laptops
 - examples
 - Oracle VirtualBox (Free - Windows/Linux/Mac)
 - VMWare Wokstation (Free - after Broadcom acquired VMWare - Windows, Linux, Mac)
 - VMWare Fusion (Mac OS-X - Need a commercial license)
 - Parallels (Mac OS-X - Need a commercial license)
- virtualization helps organization save cost in many fold
 - with less physical servers, many virtual machines(Guest OS) can be supported
 - saves cost in terms of procuring less physical servers
 - saves cost in terms of power consumption
 - saves cost in terms of Air Condition
 - saves cost in terms of Sound proofing
 - saves cost in terms of Real estate rental/leasing
- this type of virtualization is called heavy-weight
 - because each VM requires dedicated Hardware resources
 - each VM runs a fully functional OS with its own dedicated OS Kernel
 - each VM, represents one fully function Operating System

Info - Containerization

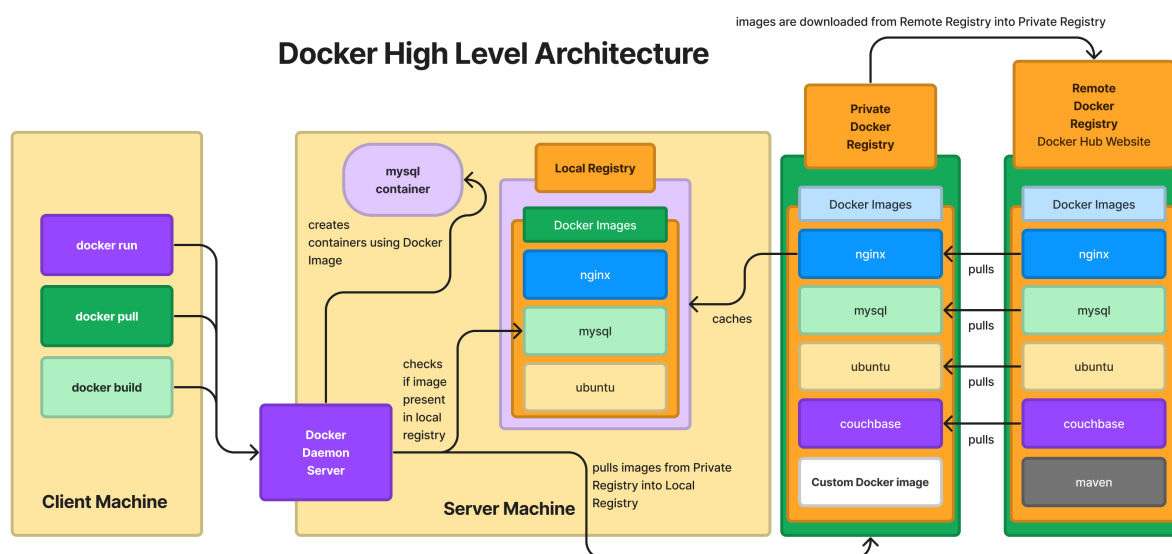
- application virtualization technology
- it is light weight, because each container represents a single application not an Operating System

- all the containers that runs on top a OS shares the Hardware resources available on the underlying OS
- containers will never be able to replace an Operating System or Virtualization
- in fact, in real-world scenarios, virtualization and containerization are used in combination
- it is a linux technology
- linux kernel features that enables containerization
 1. Namespace
 - helps isolation of containers from each other
 2. Control Groups (CGroups)
 - it helps us apply resource quota restrictions like
 - we can restrict how much RAM, disk and cpu can be utilized by a container
- container runtime
 - is a low-level software that manages container images and containers
 - it is not user-friendly, hence end-user like us never use container runtime directly
 - examples
 - runC
 - cRun
 - CRI-O
- container engine
 - is a high-level software that manages container images and containers
 - it is very user-friendly, but they depend on container runtimes internally to manage images and containers
 - examples
 - docker
 - podman

Info - Hypervisor High-Level Architecture



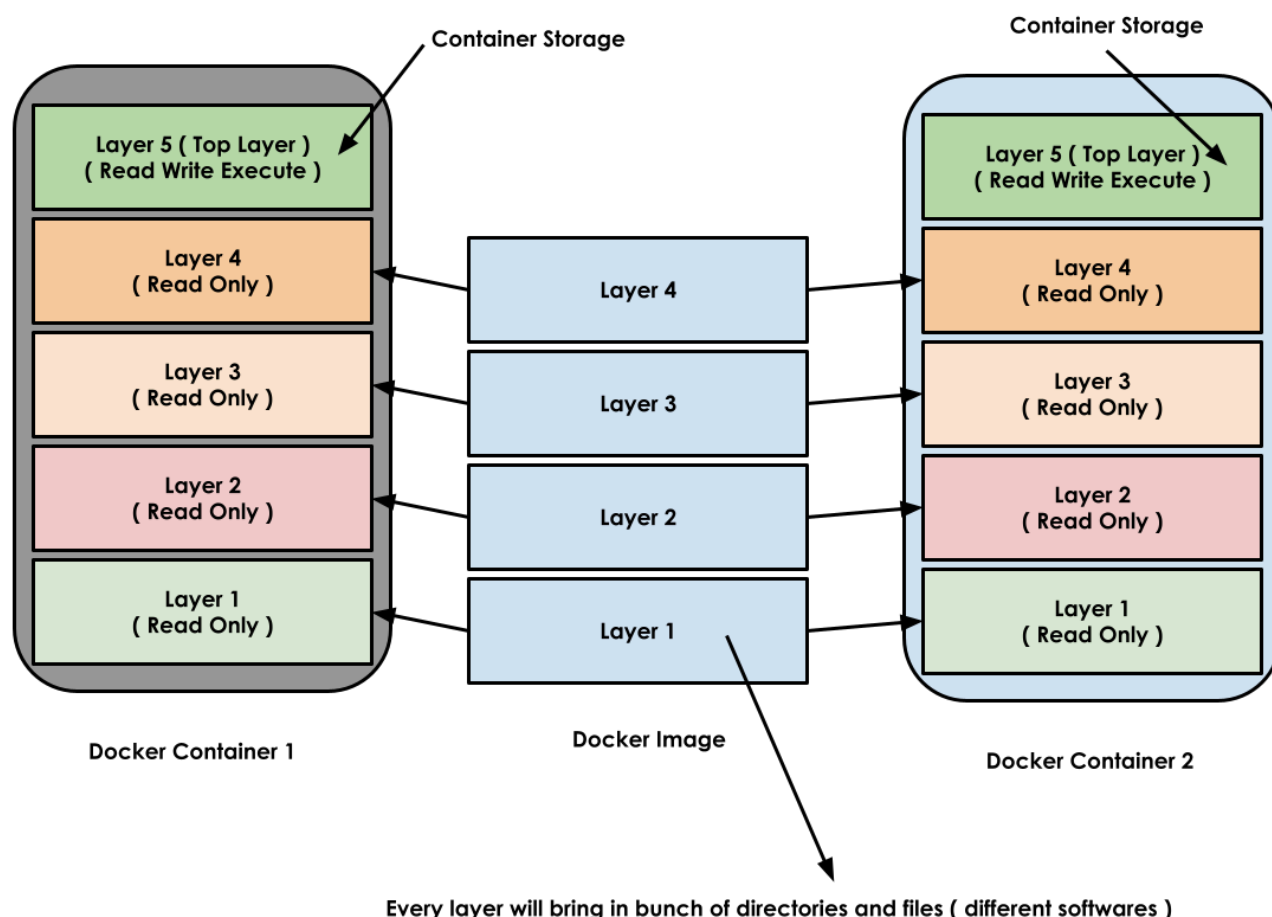
Info - Docker High-Level Architecture



Info - Docker image

- is a blueprint of a docker container
- you can imagine docker image similar to Window12.iso (OS Image) or Ubuntu24.04.iso(OS Image)
- with the help of a docker image, we can create any number of docker containers
- docker images => typically has one application and its dependencies (libraries, dependent tools, etc.,)
- docker images are conservatively built, which means it only contains bare minimum tools required to run a specific application
- every docker images gets a unique name and unique id (256-bit HASH)

Docker Image Layers to Container File System



Info - Docker Container

- is the running instance of a Docker Image (Container Image)
- each container gets a unique name and ID (SHA Hash)
- each container gets its own dedicated software defined network stack (7 OSI Layers)
- each container get its own file system (files & folders)
- each container uses about 7 namespaces
- each container gets one or more IP addresses (generally Private IP)

Info - Docker Registry

- is a collection of multiple docker images
- there are 3 types
 1. Local Docker Registry
 - is a folder created and maintained by Docker server on the same machine where it runs
 - for example, in linux distros, local docker registry is maintained under directory /var/lib/docker
 2. Remote docker registry
 - is a web site (hub.docker.com) maintained by Docker Inc and opensource community

3. Private Docker registry

- can be setup using JFrog Artifactory or Sonatype Nexus

Demo - Installing Docker community edition (Docker CE)

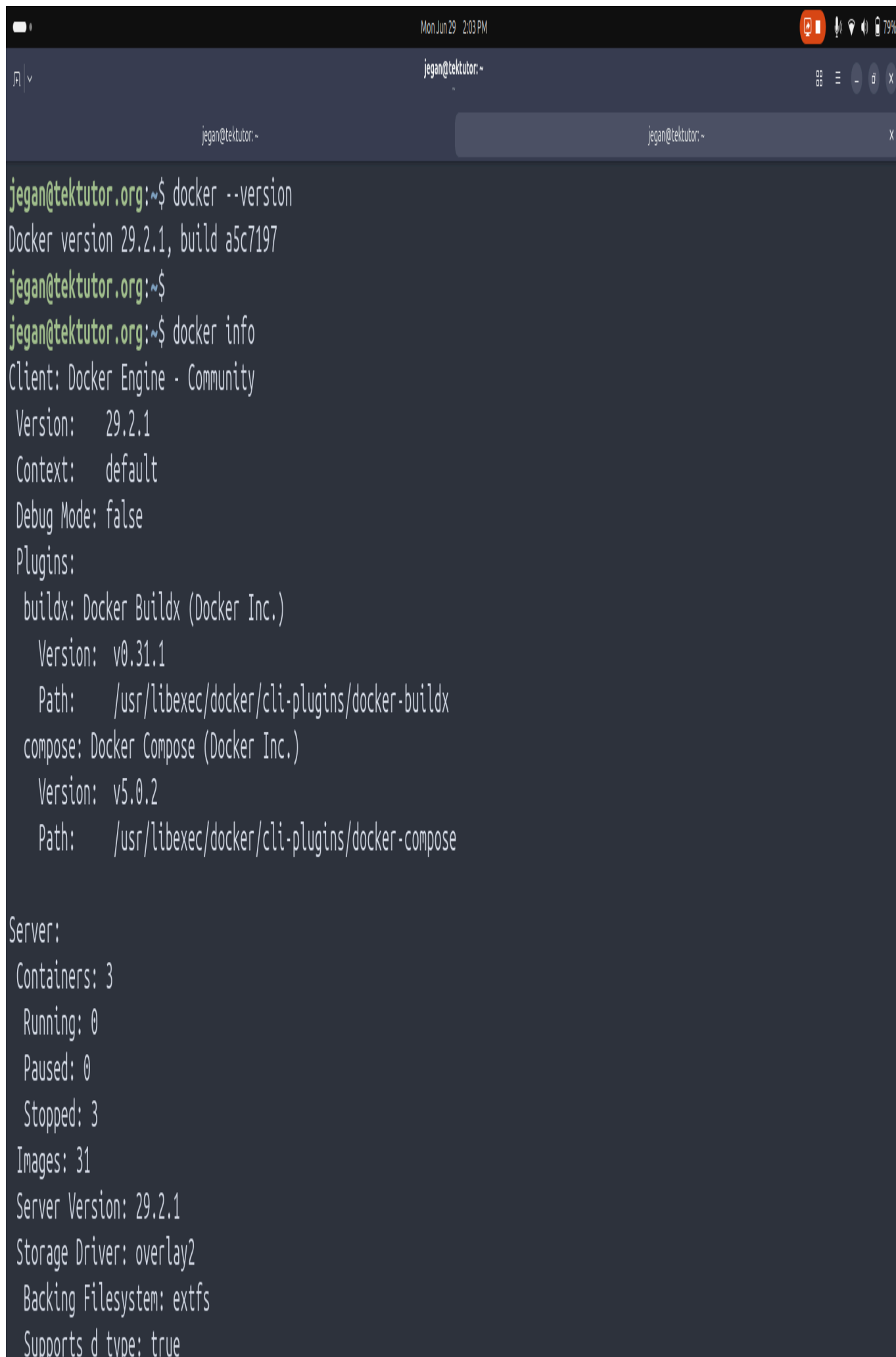
```
# Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo
"${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin -y
sudo systemctl enable docker
sudo systemctl start docker
sudo usermod -aG docker $USER
newgrp docker
```

Lab - Checking docker details

```
docker --version
docker info
```

A screenshot of a terminal window on a Linux system. The window title is 'jegan@tektutor: ~'. The terminal shows the execution of 'docker --version' and 'docker info' commands. The output of 'docker --version' is 'Docker version 29.2.1, build a5c7197'. The output of 'docker info' shows client and server information, including version 29.2.1, context 'default', and details about plugins (buildx, compose) and server status (3 containers, 31 images, overlay2 storage driver).

```
jegan@tektutor.org:~$ docker --version
Docker version 29.2.1, build a5c7197
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker info
Client: Docker Engine - Community
Version:      29.2.1
Context:      default
Debug Mode:   false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.31.1
    Path:      /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.0.2
    Path:      /usr/libexec/docker/cli-plugins/docker-compose

Server:
Containers: 3
  Running: 0
  Paused: 0
  Stopped: 3
Images: 31
Server Version: 29.2.1
Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d type: true
```

```
Using metacopy: false  
Native Overlay Diff: true  
userxattr: false  
Logging Driver: json-file  
Cgroup Driver: systemd
```

```
Mon Jun 29 2:06 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~

Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: dea7da592f5d1d2b7755e3a161be07f43fad8f75
runc version: v1.3.4-0-gd6d73eb8
init version: de40ad0
Security Options:
  apparmor
  seccomp
    Profile: builtin
  cgroupns
Kernel Version: 7.0.0-22-generic
Operating System: Ubuntu 26.04 LTS
OSType: linux
Architecture: x86_64
CPUs: 24
Total Memory: 61.23GiB
Name: tektutor.org
ID: 9554324f-6d2b-40e7-8920-5e553fa0f8a3
Docker Root Dir: /var/lib/docker
Debug Mode: false
Username: tektutor
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
```



```
Live Restore Enabled: false  
Firewall Backend: iptables
```

```
jegan@tektutor.org:~$
```

Lab - Listing docker images from your local docker registry

```
docker images
```

```
Mon Jun 29 2:07 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	3cb067eab609	8.45MB	0B	
badimage-ubuntu:1.0	5736020bd49c	992MB	0B	
buchgr/bazel-remote-cache:latest	aad72c4b8fe1	38.5MB	0B	
database:latest	27deb4ec2bb4	445MB	0B	
docker/docker-bench-security:latest	0037349aef7e	51.6MB	0B	
gaiaadm/pumba:latest	fc4eb3c7aae3	28.9MB	0B	
gcr.io/cadvisor/cadvisor:v0.49.1	c02cf39d3dba	80.8MB	0B	
gcr.io/k8s-minikube/kicbase:v0.0.50	6da180ef5035	1.37GB	0B	U
goodimage-ubuntu:1.0	5ec1158c04f9	339MB	0B	
high-throughput-app:latest	da954fb959a3	62.3MB	0B	
high-throughput-app:v1.0	da954fb959a3	62.3MB	0B	
high-throughput-app:v2.0	da954fb959a3	62.3MB	0B	
high-throughput-app:v2.1	da954fb959a3	62.3MB	0B	
legacy-app:latest	da954fb959a3	62.3MB	0B	
legacy-app:v1.0	da954fb959a3	62.3MB	0B	
legacy-app:v2.0	da954fb959a3	62.3MB	0B	
legacy-app:v2.1	da954fb959a3	62.3MB	0B	
monitoring-agent:latest	da954fb959a3	62.3MB	0B	
monitoring-agent:v1.0	da954fb959a3	62.3MB	0B	
monitoring-agent:v2.0	da954fb959a3	62.3MB	0B	
monitoring-agent:v2.1	da954fb959a3	62.3MB	0B	
myapi:latest	da954fb959a3	62.3MB	0B	
myapi:v1.0	da954fb959a3	62.3MB	0B	

```
myapi:v2.0          da954fb959a3    62.3MB    0B
myapi:v2.1          da954fb959a3    62.3MB    0B
myapp:latest        da954fb959a3    62.3MB    0B
myapp:modified      4ef0d95a33d4    62.3MB    0B
myapp:v1.0          da954fb959a3    62.3MB    0B
```

Lab - Downloading docker images from docker hub to local registry

```
docker pull hello-world:latest
docker pull ubuntu:latest

docker images
```

```
Mon Jun 29 2:09 PM
jegan@tektutor:~

jegan@tektutor.org:~$ # Download docker images from docker hub remote registry to local docker registry
jegan@tektutor.org:~$ docker pull hello-world:latest
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker pull ubuntu:latest
latest: Pulling from library/ubuntu
81e2f2053c8f: Pull complete
d1f56e4c7f2f: Pull complete
Digest: sha256:53958ec7b67c2c9355df922dd08dbf0360611f8c3cdb656875e81873db9ffdba
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker images | grep ubuntu
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
badimage-ubuntu:1.0          5736020bd49c      992MB             0B
goodimage-ubuntu:1.0         5ec1158c04f9      339MB             0B
tektutor/ubuntu:1.0          0ffe51b3f464      1.08GB            0B
ubuntu:22.04                 86f1a8d7b38e      78.1MB            0B
ubuntu:latest                5a4c6b929c57      100MB             0B
jegan@tektutor.org:~$
```



Lab - Deleting a docker image from your local registry

```
docker images | grep hello  
docker rmi hello-world:latest  
docker images | grep hello
```

```
Mon Jun 29 2:21 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org:~$ docker images | grep hello
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
hello-world:latest          e2ac70e7319a      10.1kB           0B
tektutor/hello-microservice:1.0  540ee094e2c8      412MB           0B
tektutor/hello-ms:1.0         c588ed117d2c      411MB           0B
jegan@tektutor.org:~$
jegan@tektutor.org:~$ # Deleting a docker image from your local registry
jegan@tektutor.org:~$ docker rmi hello-world:latest
Untagged: hello-world:latest
Untagged: hello-world@sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Deleted: sha256:e2ac70e7319a02c5a477f5825259bd118b94e8b02c279c67afa63adab6d8685b
Deleted: sha256:897b3f2a7c1bc2f3d02432f7892fe31c6272c521ad4d70257df624504a3238b4
jegan@tektutor.org:~$ docker images | grep hello
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
tektutor/hello-microservice:1.0  540ee094e2c8      412MB           0B
tektutor/hello-ms:1.0         c588ed117d2c      411MB           0B
jegan@tektutor.org:~$
```

Lab - Creating a container in the interactive(foreground) mode

From terminal 1

```
docker run -it --name ubuntu1-jegan --hostname ubuntu1-jegan ubuntu:latest  
/bin/bash  
hostname  
hostname -i  
ls  
exit
```

List all running containers from second terminal

```
docker ps | grep jegan
```

List all running containers you created

```
docker ps -a | grep jegan
```

```
Mon Jun 29 3:03 PM
jegan@tektutor: ~/custom-docker-image
jegan@tektutor: ~/custom-docker-image

jegan@tektutor.org:~/custom-docker-image$ docker images | grep ubuntu:26.04
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ubuntu:26.04          5a4c6b929c57          100MB          0B
jegan@tektutor.org:~/custom-docker-image$
jegan@tektutor.org:~/custom-docker-image$ docker run -it --name ubuntu1 --hostname ubuntu1 ubuntu:latest /bin/bash
root@ubuntu1:/#
root@ubuntu1:/# hostname
ubuntu1
root@ubuntu1:/# hostname -i
172.17.0.2
root@ubuntu1:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@ubuntu1:/# exit
exit
jegan@tektutor.org:~/custom-docker-image$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
jegan@tektutor.org:~/custom-docker-image$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
8118792f4e8e   ubuntu:latest   "/bin/bash"   19 seconds ago   Exited (0) 7 seconds ago   ubuntu1
a064e9819ff3   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr..."   27 hours ago   Exited (130) 23 hours ago   minikube
7cdbf105ddc8   30ba44506a6d   "/bin/bash"   10 days ago   Exited (137) 9 days ago   ubuntu2
jegan@tektutor.org:~/custom-docker-image$
```




Lab - Create a custom docker image

In your home directory, create a folder something like custom-docker-image

```
cd ~/
mkdir -p custom-docker-image
cd custom-docker-image
touch Dockerfile
```

Add the below content to your Dockerfile, you can run gedit to open windows notepad like editor

```
FROM ubuntu:26.04

RUN apt update && apt install -y net-tools iputils-ping
RUN apt update && apt install -y default-jdk maven
```

Build your custom docker image

```
cd ~/custom-docker-image
docker build -t ubuntu-jegan:1.0 .
docker images | grep jegan
```

Create a container using your custom docker image

```
docker run -dit --name c1-jegan --hostname c1-jegan ubuntu-jegan:1.0
/bin/bash
docker exec -it c1-jegan /bin/bash
hostname
hostname -i
ping 8.8.8.8
ifconfig
ls
exit
```

List and see the container should still run because we created this container in background

```
docker ps
```

Lab - Stopping a running container

```
docker ps | grep c1-jegan  
docker stop c1-jegan  
docker ps | grep c1-jegan  
docker ps -a | grep c1-jegan
```

Lab - Start an exited container

```
docker ps -a | grep c1-jegan  
docker start c1-jegan  
docker ps | grep c1-jegan
```

Lab - Restart a running container

```
docker ps a | grep c1-jegan  
docker restart c1-jegan  
docker ps | grep c1-jegan
```

Lab - Deleting a running container

In order to delete a running container gracefully, you must stop it first

```
docker stop c1-jegan  
docker rm c1-jegan  
docker ps -a
```

In order to delete a running container forcibly

```
docker rm -f c1-jegan  
docker ps -a
```

Lab - Rename a container

```
docker run -dit ubuntu:latest /bin/bash  
docker ps a
```

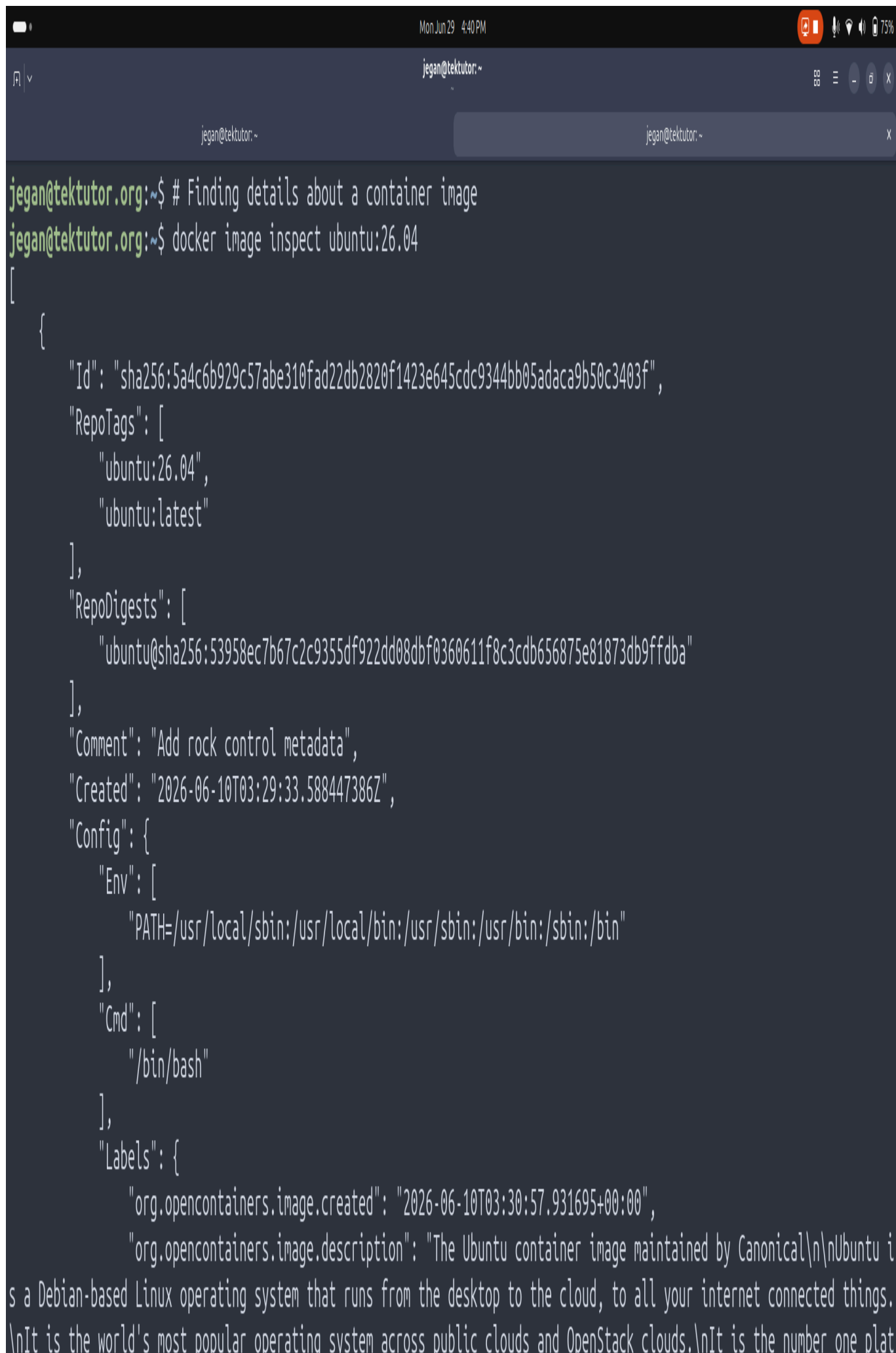
```
docker rename zealous_montalcini c2-jegan  
docker ps
```

```
Mon Jun 29 4:33 PM
jegan@tektutor:~
jegan@tektutor:~
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
jegan@tektutor.org:~$ docker run -dit --name c1-jegan --hostname c1-jegan ubuntu:latest /bin/bash
5a8c2b4e46924008351d1d41c4bcd51544b0b262c69ae46573729ad094ca2f9f
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker run -dit ubuntu:latest /bin/bash
f93b087d7f6d8f27d5dc2248832f4809e82375fe7702ff389d1e6a3e8ccdf496
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
f93b087d7f6d   ubuntu:latest   "/bin/bash"   5 seconds ago   Up 4 seconds   zealous_montalcini
5a8c2b4e4692   ubuntu:latest   "/bin/bash"   2 minutes ago   Up 2 minutes   c1-jegan
jegan@tektutor.org:~$ docker rename zealous_montalcini c2-jegan
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
f93b087d7f6d   ubuntu:latest   "/bin/bash"   26 seconds ago   Up 25 seconds   c2-jegan
5a8c2b4e4692   ubuntu:latest   "/bin/bash"   2 minutes ago   Up 2 minutes   c1-jegan
jegan@tektutor.org:~$ docker exec -it c2-jegan /bin/bash
root@f93b087d7f6d:/# exit
exit
jegan@tektutor.org:~$ hostname
tektutor.org
jegan@tektutor.org:~$ docker exec -it c2-jegan hostname
f93b087d7f6d
jegan@tektutor.org:~$ docker exec -it c2-jegan hostname -i
172.17.0.3
jegan@tektutor.org:~$
```



Lab - Finding details about a container image

```
docker image inspect ubuntu:26.04
```



A terminal window titled 'jegan@tektutor: ~' displays the command 'docker image inspect ubuntu:26.04'. The output is a JSON array containing details about the container image. The terminal interface includes a top status bar with the date 'Mon Jun 29 4:40 PM' and system icons, and a window title bar with standard Linux window controls.

```
jegan@tektutor.org:~$ # Finding details about a container image
jegan@tektutor.org:~$ docker image inspect ubuntu:26.04
[
  {
    "Id": "sha256:5a4c6b929c57abe310fad22db2820f1423e645cdc9344bb05adaca9b50c3403f",
    "RepoTags": [
      "ubuntu:26.04",
      "ubuntu:latest"
    ],
    "RepoDigests": [
      "ubuntu@sha256:53958ec7b67c2c9355df922dd08dbf0360611f8c3cdb656875e81873db9ffdba"
    ],
    "Comment": "Add rock control metadata",
    "Created": "2026-06-10T03:29:33.588447386Z",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Cmd": [
        "/bin/bash"
      ],
      "Labels": {
        "org.opencontainers.image.created": "2026-06-10T03:30:57.931695+00:00",
        "org.opencontainers.image.description": "The Ubuntu container image maintained by Canonical\n\nUbuntu i
a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
\nIt is the world's most popular operating system across public clouds and OpenStack clouds.\nIt is the number one plat
```

```
form for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.\nFast, secure and simple, Ubuntu powers millions of PCs worldwide.\n",  
    "org.opencontainers.image.title": "ubuntu",  
    "org.opencontainers.image.version": "26.04"  
}
```

```
Mon Jun 29 4:42 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~

},
  "Architecture": "amd64",
  "Os": "linux",
  "Size": 100154952,
  "GraphDriver": {
    "Data": {
      "LowerDir": "/var/lib/docker/overlay2/6cf3ac66468bf0f3ea9d4349f47a5d70e90f6e10508e4a6ded47024c3fa4a4cb/
diff",
      "MergedDir": "/var/lib/docker/overlay2/9a98a26589c570a1d29fc482219e1c1d455286d65d619a9003602668b7d97adb
/merged",
      "UpperDir": "/var/lib/docker/overlay2/9a98a26589c570a1d29fc482219e1c1d455286d65d619a9003602668b7d97adb/
diff",
      "WorkDir": "/var/lib/docker/overlay2/9a98a26589c570a1d29fc482219e1c1d455286d65d619a9003602668b7d97adb/w
ork"
    }
  },
  "Name": "overlay2"
},
"RootFS": {
  "Type": "layers",
  "Layers": [
    "sha256:e8c084c1b320c172e8be941d735d77298e49986014213a8282c9b533ee216a61",
    "sha256:ab0ae19b58df847c6f0e4beb31f759a4dac2cc42288a6bda6f467eea4584c541"
  ]
},
"Metadata": {
  "LastTagTime": "0001-01-01T00:00:00Z"
}
```



```
}  
}  
]  
jegan@tektutor.org:~$
```

Lab - Finding details about a container

```
docker container inspect c1-jegan  
docker inspect c1-jegan
```

```
Mon Jun 29 4:43 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE          COMMAND          CREATED          STATUS          PORTS          NAMES
f93b087d7f6d   ubuntu:latest  "/bin/bash"     39 minutes ago  Up 39 minutes           c2-jegan
5a8c2b4e4692   ubuntu:latest  "/bin/bash"     41 minutes ago  Up 41 minutes           c1-jegan
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker container inspect c1-jegan
[
  {
    "Id": "5a8c2b4e46924008351d1d41c4bcd51544b0b262c69ae46573729ad094ca2f9f",
    "Created": "2026-06-29T10:31:37.081139843Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 53001,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-06-29T10:31:37.116359236Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:5a4c6b929c57abe310fad22db2820f1423e645cdc9344bb05adaca9b50c3403f",
```

```
    "ResolvConfPath": "/var/lib/docker/containers/5a8c2b4e46924008351d1d41c4bcd51544b0b262c69ae46573729ad094ca2f9f/  
    resolv.conf",  
    "HostnamePath": "/var/lib/docker/containers/5a8c2b4e46924008351d1d41c4bcd51544b0b262c69ae46573729ad094ca2f9f/ho  
    stname",  
    "HostsPath": "/var/lib/docker/containers/5a8c2b4e46924008351d1d41c4bcd51544b0b262c69ae46573729ad094ca2f9f/hosts
```

A terminal window with a dark background and light gray text. The window title bar shows 'Mon Jun 29 4:43 PM' and system icons on the right. The terminal prompt is 'jegan@tektutor: ~'. The output is a JSON configuration for Docker network settings, including 'org.opencontainers.image.version', 'NetworkSettings', 'SandboxID', 'SandboxKey', 'Ports', 'Networks', and 'bridge' details like 'IPAMConfig', 'Links', 'Aliases', 'DriverOpts', 'GwPriority', 'NetworkID', 'EndpointID', 'Gateway', 'IPAddress', 'MacAddress', 'IPPrefixLen', 'IPv6Gateway', 'GlobalIPv6Address', 'GlobalIPv6PrefixLen', and 'DNSNames'.

```

    "org.opencontainers.image.version": "26.04"
  },
  "NetworkSettings": {
    "SandboxID": "e451ed67dfc919c041ec85159dec162814022b6125accb5e014607192133c5fe",
    "SandboxKey": "/var/run/docker/netns/e451ed67dfc9",
    "Ports": {},
    "Networks": {
      "bridge": {
        "IPAMConfig": null,
        "Links": null,
        "Aliases": null,
        "DriverOpts": null,
        "GwPriority": 0,
        "NetworkID": "4e636f92ca261130ad5d421a9e5c7e8f8da49905a8b9482754d831e7103d7119",
        "EndpointID": "efea4f0d870a331996cdc9090f82eb3696b3cc693429e61485aec2a96e263bd9",
        "Gateway": "172.17.0.1",
        "IPAddress": "172.17.0.2",
        "MacAddress": "6a:66:ac:7f:54:1c",
        "IPPrefixLen": 16,
        "IPv6Gateway": "",
        "GlobalIPv6Address": "",
        "GlobalIPv6PrefixLen": 0,
        "DNSNames": null
      }
    }
  }
}
```



Lab - Finding IP address of a container

```
docker inspect c1-jegan | grep IPA
docker inspect -f {{.NetworkSettings.IPAddress}} c1-jegan
docker exec -it c1-jegan hostname -i
```

```
Mon Jun 29 4:45 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~

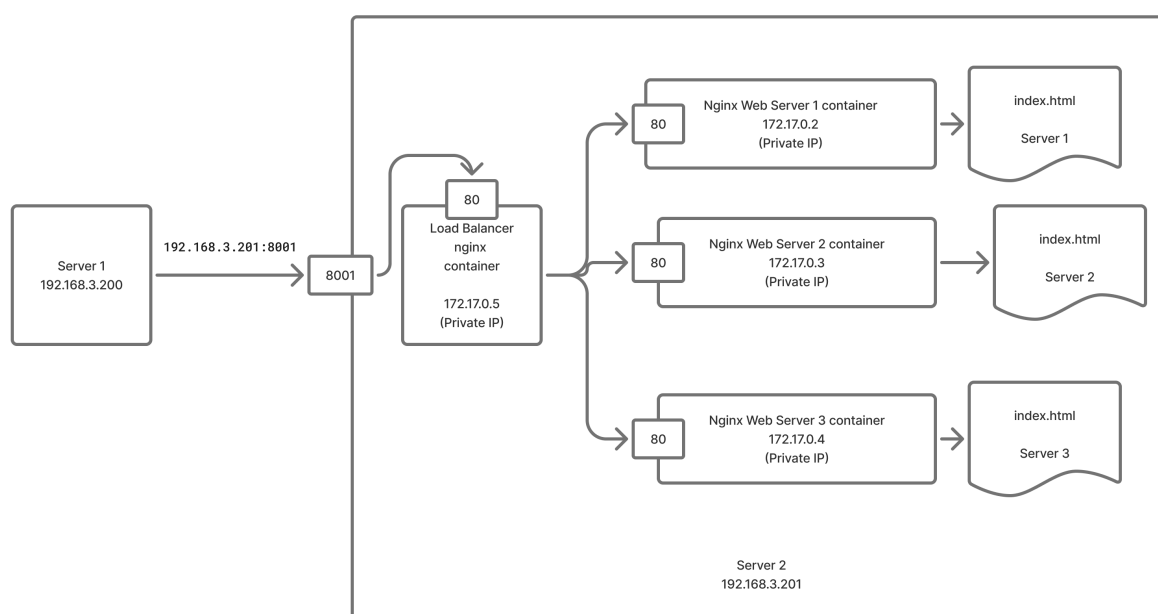
"SandboxID": "e451ed67dfc919c041ec85159dec162814022b6125accb5e014607192133c5fe",
"SandboxKey": "/var/run/docker/netns/e451ed67dfc9",
"Ports": {},
"Networks": {
  "bridge": {
    "IPAMConfig": null,
    "Links": null,
    "Aliases": null,
    "DriverOpts": null,
    "GwPriority": 0,
    "NetworkID": "4e636f92ca261130ad5d421a9e5c7e8f8da49905a8b9482754d831e7103d7119",
    "EndpointID": "efea4f0d870a331996cdc9090f82eb3696b3cc693429e61485aec2a96e263bd9",
    "Gateway": "172.17.0.1",
    "IPAddress": "172.17.0.2",
    "MacAddress": "6a:66:ac:7f:54:1c",
    "IPPrefixLen": 16,
    "IPv6Gateway": "",
    "GlobalIPv6Address": "",
    "GlobalIPv6PrefixLen": 0,
    "DNSNames": null
  }
}
}
}
]

jegan@tektutor.org:~$ docker inspect -f {{.NetworkSettings.Networks.bridge.IPAddress}} c1-jegan
```

```
172.17.0.2
jegan@tektutor.org:~$ docker inspect -f {{.NetworkSettings.SandboxKey}} c1-jegan
/var/run/docker/netns/e451ed67dfc9
jegan@tektutor.org:~$
```

Lab - Setup a Load balancer using containers

Docker - Port Forwarding



Let's create 3 web server containers using nginx:latest image from Docker Hub website

```
docker run -d --name nginx1-jegan --hostname nginx1-jegan nginx:latest
docker run -d --name nginx2-jegan --hostname nginx2-jegan nginx:latest
docker run -d --name nginx3-jegan --hostname nginx3-jegan nginx:latest
```

List all running containers

```
docker ps
```

Find the IP addresses of the 3 webserver containers

```
docker inspect nginx1-jegan | grep IPA  
docker inspect nginx2-jegan | grep IPA  
docker inspect nginx3-jegan | grep IPA
```

See if you are able to access the web pages from those web server containers

```
curl http://172.17.0.2:80  
curl http://172.17.0.3:80  
curl http://172.17.0.4:80
```



```
Mon Jun 29 5:04 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org:~$ docker run -d --name nginx1-jegan --hostname nginx1-jegan nginx:latest
8d1751568faa2cb0c557d06271334a5a38632b2029d779c7b3db23ac1fa518b0
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker run -d --name nginx2-jegan --hostname nginx2-jegan nginx:latest
86f7557ff0f3dfb27d39a741dc29053e96809a910ae43d39bd6cc7fbc05b340
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker run -d --name nginx3-jegan --hostname nginx3-jegan nginx:latest
f43aac8a663087ed41edfaede99ee6d92f74f10c9a9ced6933ce208a9a13330c
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
f43aac8a6630   nginx:latest   "/docker-entrypoint..." 4 seconds ago  Up 3 seconds  80/tcp       nginx3-jegan
86f7557ff0f3   nginx:latest   "/docker-entrypoint..." 15 seconds ago  Up 14 seconds 80/tcp       nginx2-jegan
8d1751568faa   nginx:latest   "/docker-entrypoint..." 23 seconds ago  Up 23 seconds 80/tcp       nginx1-jegan
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker inspect nginx1-jegan | grep IPA
    "IPAMConfig": null,
    "IPAddress": "172.17.0.2",
jegan@tektutor.org:~$ docker inspect nginx2-jegan | grep IPA
    "IPAMConfig": null,
    "IPAddress": "172.17.0.3",
jegan@tektutor.org:~$ docker inspect nginx3-jegan | grep IPA
    "IPAMConfig": null,
    "IPAddress": "172.17.0.4",
jegan@tektutor.org:~$
jegan@tektutor.org:~$ curl http://172.17.0.2:80
```

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Welcome to nginx!</title>  
<style>
```

```
Mon Jun 29 5:08 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~$ docker inspect nginx3-jegan | grep IPA
    "IPAMConfig": null,
    "IPAddress": "172.17.0.4",
jegan@tektutor.org:~$
jegan@tektutor.org:~$ curl http://172.17.0.2:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
```

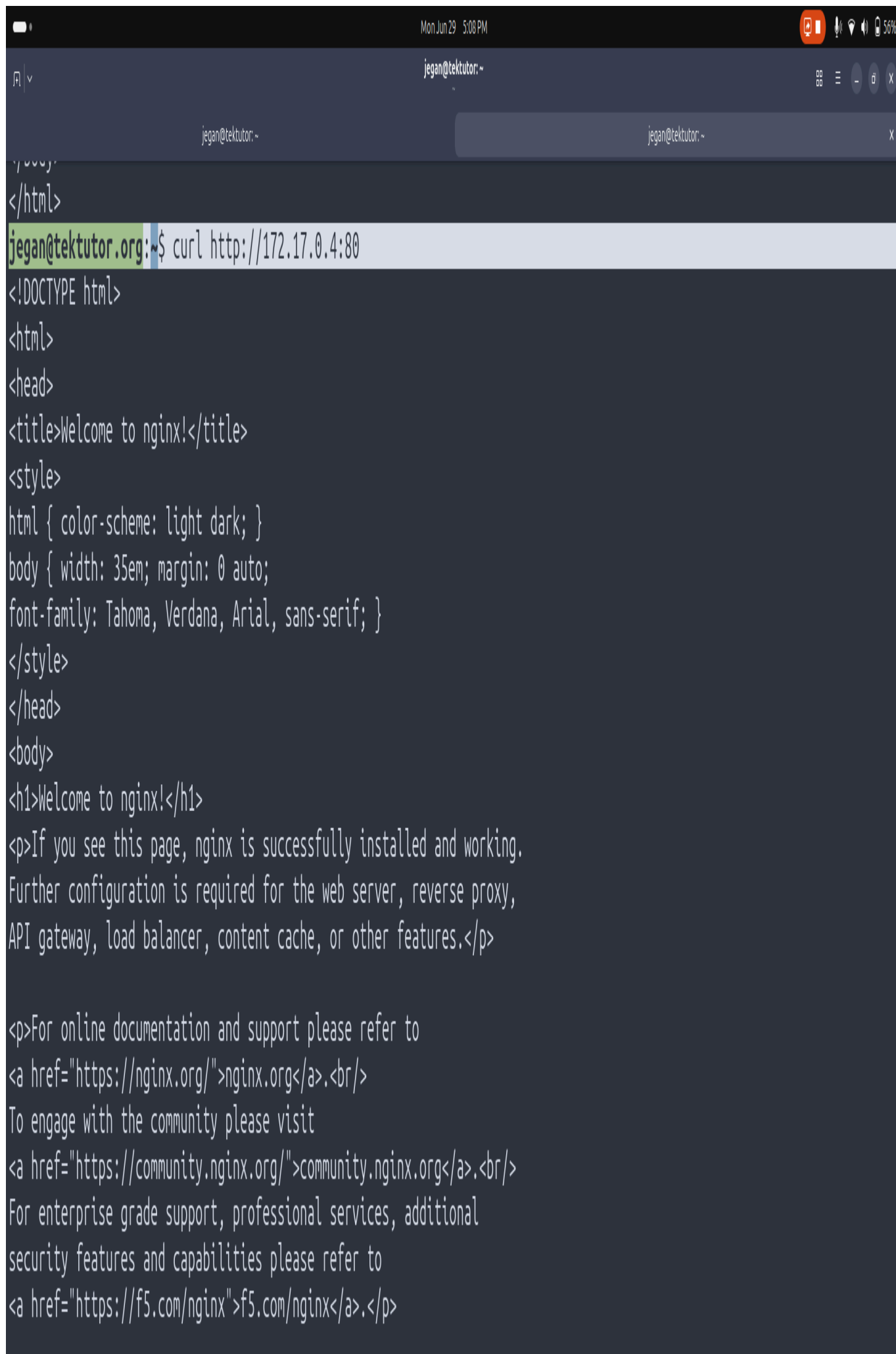
For enterprise grade support, professional services, additional

security features and capabilities please refer to

f5.com/nginx.

Thank you for using nginx.

`</body>`

A terminal window with a dark background. The title bar shows 'Mon Jun 29 5:08 PM' and system icons on the right. The terminal shows a prompt 'jegan@tektutor: ~' followed by a command 'curl http://172.17.0.4:80'. The output is an HTML document. The first line is '<!DOCTYPE html>'. The second line is '<html>'. The third line is '<head>'. The fourth line is '<title>Welcome to nginx!</title>'. The fifth line is '<style>'. The sixth line is 'html { color-scheme: light dark; }'. The seventh line is 'body { width: 35em; margin: 0 auto;'. The eighth line is 'font-family: Tahoma, Verdana, Arial, sans-serif; }'. The ninth line is '</style>'. The tenth line is '</head>'. The eleventh line is '<body>'. The twelfth line is '<h1>Welcome to nginx!</h1>'. The thirteenth line is '<p>If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.</p>'. The fourteenth line is '<p>For online documentation and support please refer to'. The fifteenth line is 'nginx.org.
'. The sixteenth line is 'To engage with the community please visit'. The seventeenth line is 'community.nginx.org.
'. The eighteenth line is 'For enterprise grade support, professional services, additional security features and capabilities please refer to'. The nineteenth line is 'f5.com/nginx.</p>'.

```
Mon Jun 29 5:08 PM
jegan@tektutor: ~
jegan@tektutor: ~$ curl http://172.17.0.4:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
```

```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
jegan@tektutor.org:~$ <!DOCTYPE html>
<html>
```

Now, let's create the load balancer container with port-forward(to make it accessible outside the system where the lb container is running)

```
docker run -d --name lb-jegan --hostname lb-jegan -p 8080:80 nginx:latest
```

In case you wish docker to identify an available port on your lab machine and forward to that port

```
docker rm -f lb-jegan
docker run -d --name lb-jegan --hostname lb-jegan -P nginx:latest
```

```
Mon Jun 29 5:41 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org:~$ docker run -d --name lb-jegan --hostname lb-jegan -P nginx:latest
a40da94fab0d9374e845493733a4c5578a064b7581220db10e3c3e5d5646dfdf
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
a40da94fab0d	nginx:latest	"/docker-entrypoint..."	2 seconds ago	Up 2 seconds	0.0.0.0:32769->80/tcp, [::]:32769->80/tcp	lb-jegan
f43aac8a6630	nginx:latest	"/docker-entrypoint..."	35 minutes ago	Up 35 minutes	80/tcp	nginx3-jegan
86f7557ff0f3	nginx:latest	"/docker-entrypoint..."	35 minutes ago	Up 35 minutes	80/tcp	nginx2-jegan
8d1751568faa	nginx:latest	"/docker-entrypoint..."	36 minutes ago	Up 36 minutes	80/tcp	nginx1-jegan

```
jegan@tektutor.org:~$
```



We need to copy the nginx.conf file from the lb-jegan container to configure it work like a load balancer

```
docker cp lb-jegan:/etc/nginx/nginx.conf .  
cat nginx.conf
```



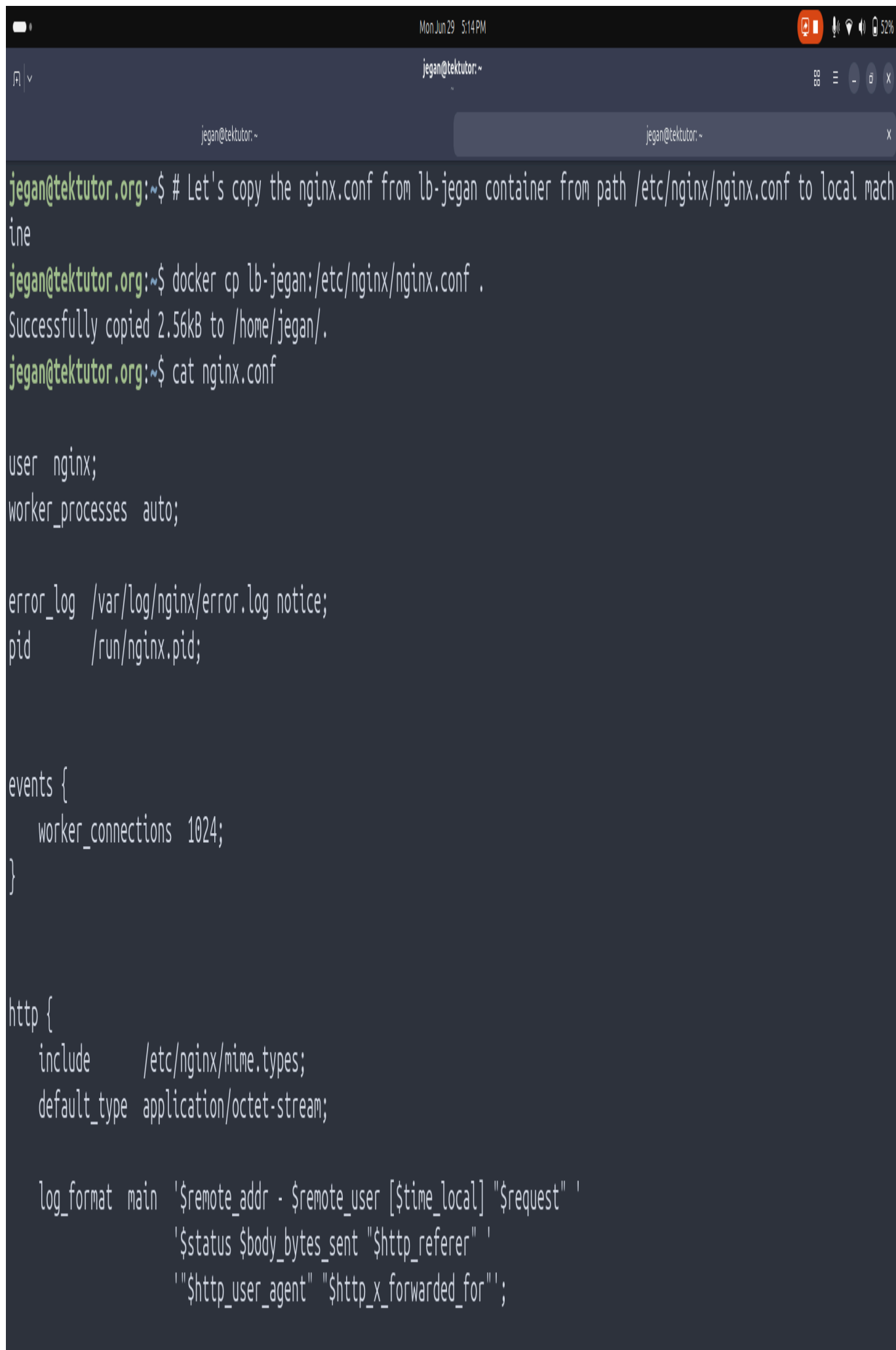
```
Mon Jun 29 5:12 PM
jegan@tektutor: ~ -- docker exec -it lb-jegan /bin/sh

jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker run -d --name lb-jegan --hostname lb-jegan -p 8080:80 nginx:latest
00b9192b12b8aa2e8266d8f371af5aac67f9e330ca12bb11f199d14f480abc4b
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
00b9192b12b8   nginx:latest   "/docker-entrypoint...." 2 seconds ago  Up 1 second   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   lb-jegan
f43aac8a6630   nginx:latest   "/docker-entrypoint...." 8 minutes ago  Up 8 minutes   80/tcp
nginx3-jegan
86f7557ff0f3   nginx:latest   "/docker-entrypoint...." 8 minutes ago  Up 8 minutes   80/tcp
nginx2-jegan
8d1751568faa   nginx:latest   "/docker-entrypoint...." 8 minutes ago  Up 8 minutes   80/tcp
nginx1-jegan
jegan@tektutor.org:~$
jegan@tektutor.org:~$ # Let's find the nginx.conf inside the lb-jegan container
jegan@tektutor.org:~$ docker exec -it lb-jegan /bin/sh
# hostname
lb-jegan
# cd /etc/nginx
# ls
conf.d  fastcgi_params  mime.types  modules  nginx.conf  scgi_params  uwsgi_params
# cat nginx.conf

user  nginx;
```

```
worker_processes auto;

error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;
```



Mon Jun 29 5:14 PM

jegan@tektutor: ~

jegan@tektutor: ~

jegan@tektutor.org:~\$ # Let's copy the nginx.conf from lb-jegan container from path /etc/nginx/nginx.conf to local machine

jegan@tektutor.org:~\$ docker cp lb-jegan:/etc/nginx/nginx.conf .

Successfully copied 2.56kB to /home/jegan/.

jegan@tektutor.org:~\$ cat nginx.conf

```
user nginx;
worker_processes auto;

error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

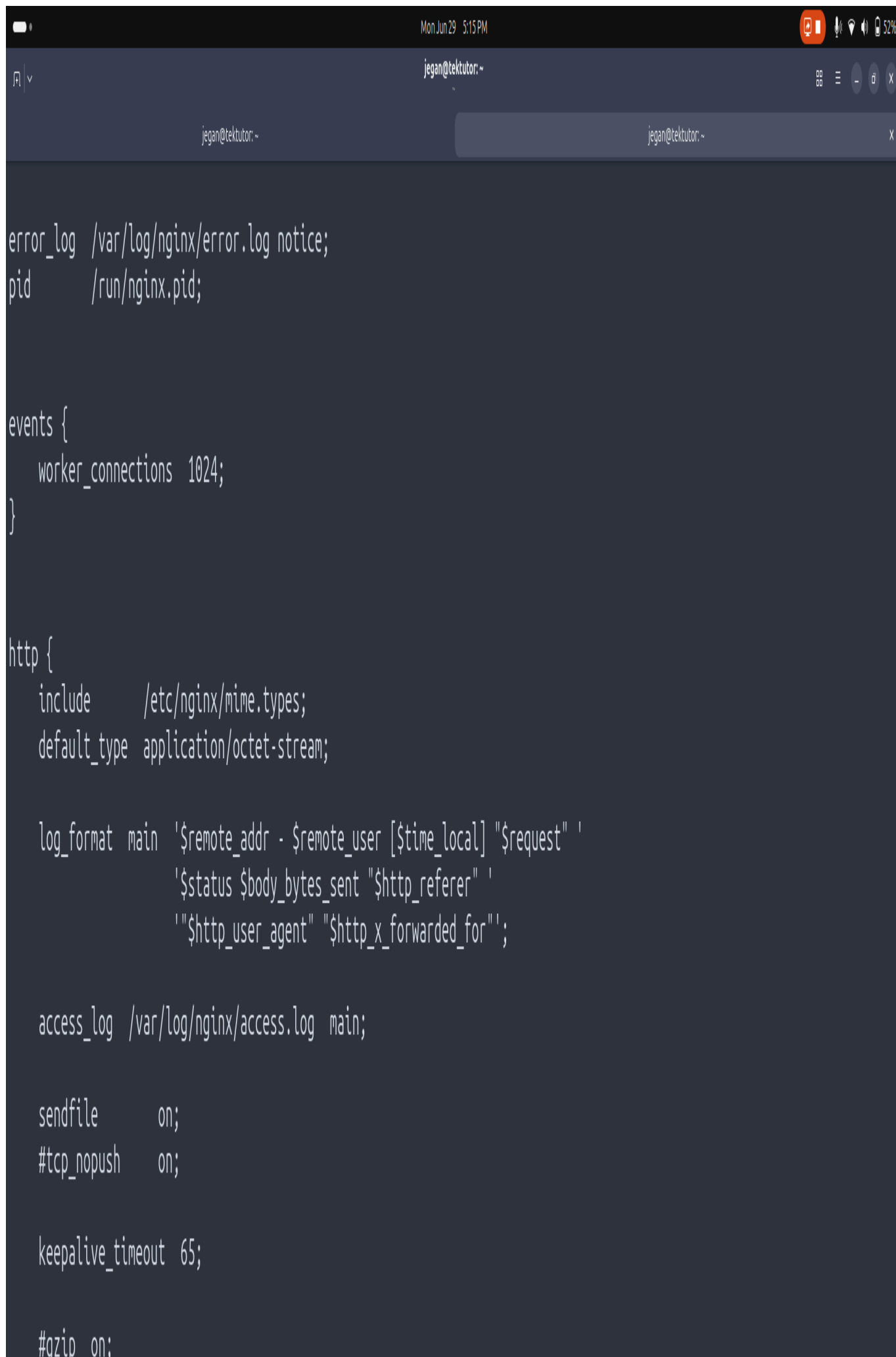
http {
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';
```

```
access_log /var/log/nginx/access.log main;
```

```
sendfile on;
```

```
#tcp_nopush on;
```

A terminal window with a dark background. The title bar at the top shows 'Mon Jun 29 5:15 PM' and system icons on the right. The terminal has a single tab titled 'jegan@tektutor: ~'. The prompt is 'jegan@tektutor: ~'. The text being entered is an nginx configuration snippet. The configuration includes error_log, pid, events, http, log_format, access_log, sendfile, #tcp_nopush, keepalive_timeout, and #gzip. The text is as follows:

```
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

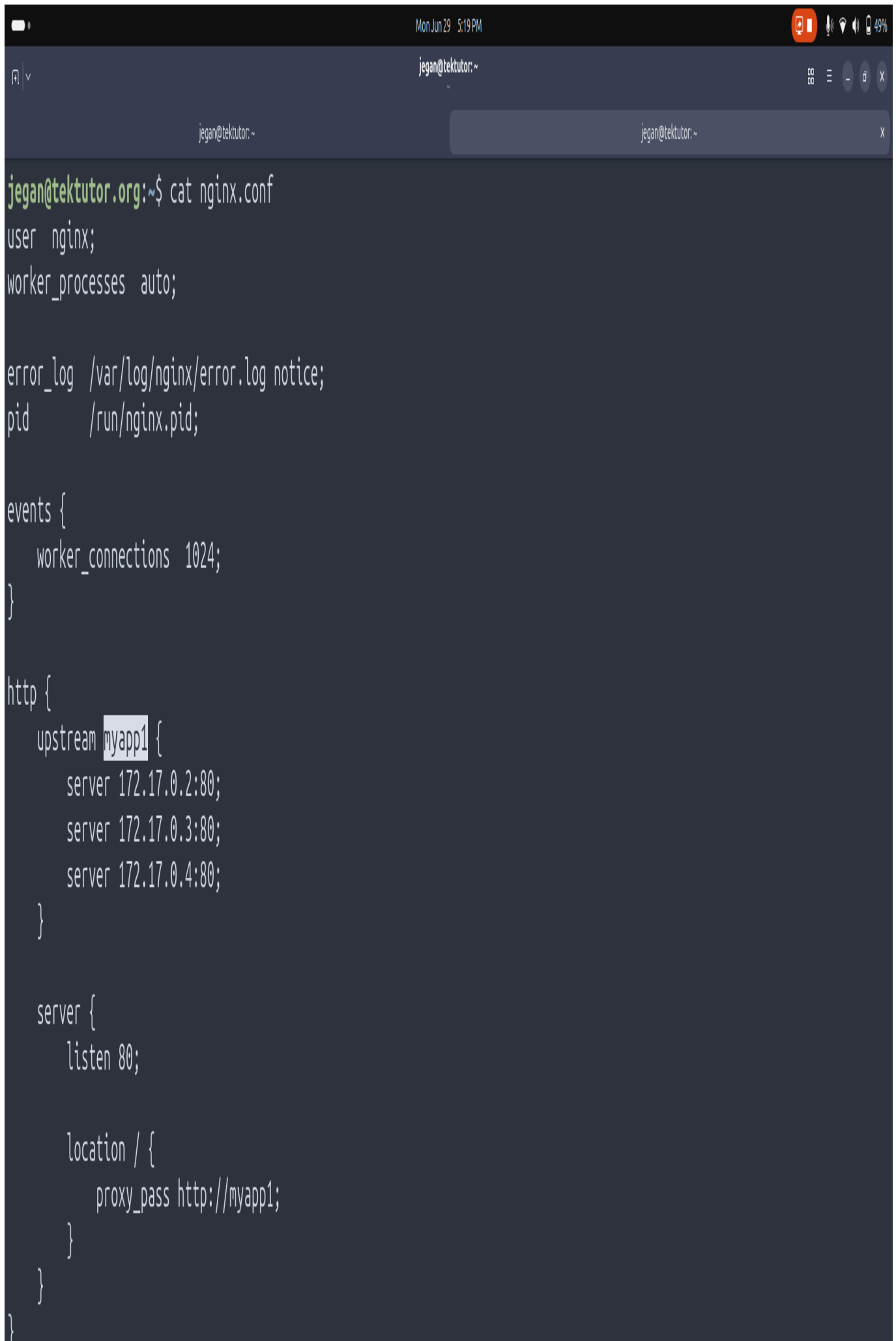
    sendfile on;
    #tcp_nopush on;

    keepalive_timeout 65;

    #gzip on;
```

```
include /etc/nginx/conf.d/*.conf;  
}  
jegan@tektutor.org:~$
```

You need find the IP addresses of your nginx web server containers and update the nginx.conf file as shown below



A terminal window titled 'jegan@tektutor: ~' is shown. The window has a dark background and a light-colored title bar. The terminal displays the output of the command 'cat nginx.conf'. The output shows the configuration for the nginx web server, including user, worker processes, error log, pid, events, and http sections. The http section includes an upstream block for 'myapp1' and a server block listening on port 80, proxying requests to 'http://myapp1'.

```
jegan@tektutor.org:~$ cat nginx.conf
user nginx;
worker_processes auto;

error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    upstream myapp1 {
        server 172.17.0.2:80;
        server 172.17.0.3:80;
        server 172.17.0.4:80;
    }

    server {
        listen 80;

        location / {
            proxy_pass http://myapp1;
        }
    }
}
```



You need to update the nginx.conf file with your web server IPs

```
user  nginx;
worker_processes  auto;

error_log  /var/log/nginx/error.log notice;
pid        /run/nginx.pid;

events {
    worker_connections  1024;
}

http {
    upstream myapp1 {
        server 172.17.0.2:80;
        server 172.17.0.3:80;
        server 172.17.0.4:80;
    }

    server {
        listen 80;

        location / {
            proxy_pass http://myapp1;
        }
    }
}
```

We need to copy this updated nginx.conf file from our local machine to the lb-jegan container

```
docker cp nginx.conf lb-jegan:/etc/nginx/nginx.conf
```

We need to restart the lb-jegan container to apply the config changes

```
docker restart lb-jegan
```



```
Mon Jun 29 5:29 PM
jegan@tektutor: ~
jegan@tektutor: ~
user nginx;
worker_processes auto;

error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    upstream myapp1 {
        server 172.17.0.2:80;
        server 172.17.0.3:80;
        server 172.17.0.4:80;
    }

    server {
        listen 80;

        location / {
            proxy_pass http://myapp1;
        }
    }
}

jegan@tektutor.org:~$ docker cp nginx.conf lb-jegan:/etc/nginx/nginx.conf
Successfully copied 2.05kB to lb-jegan:/etc/nginx/nginx.conf
jegan@tektutor.org:~$ docker restart lb-jegan
lb-jegan
jegan@tektutor.org:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
00b9192b12b8 nginx:latest "/docker-entrypoint..." 17 minutes ago Up 4 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp lb-jegan
f43aac8a6630 nginx:latest "/docker-entrypoint..." 25 minutes ago Up 25 minutes 80/tcp nginx3-jegan
86f7557ff0f3 nginx:latest "/docker-entrypoint..." 25 minutes ago Up 25 minutes 80/tcp nginx2-jegan
8d1751568faa nginx:latest "/docker-entrypoint..." 26 minutes ago Up 26 minutes 80/tcp nginx1-jegan
jegan@tektutor.org:~$
```

Make sure the lb-jegan container is running after our config changes

```
docker ps
```

Let's customize the web page in nginx1-jegan, nginx2-jegan and nginx3-jegan web server containers

```
echo "<h1>Web Server 1</h1>" > index.html
docker cp index.html nginx1-jegan:/usr/share/nginx/html/index.html

echo "<h1>Web Server 2</h1>" > index.html
docker cp index.html nginx2-jegan:/usr/share/nginx/html/index.html

echo "<h1>Web Server 3</h1>" > index.html
docker cp index.html nginx3-jegan:/usr/share/nginx/html/index.html

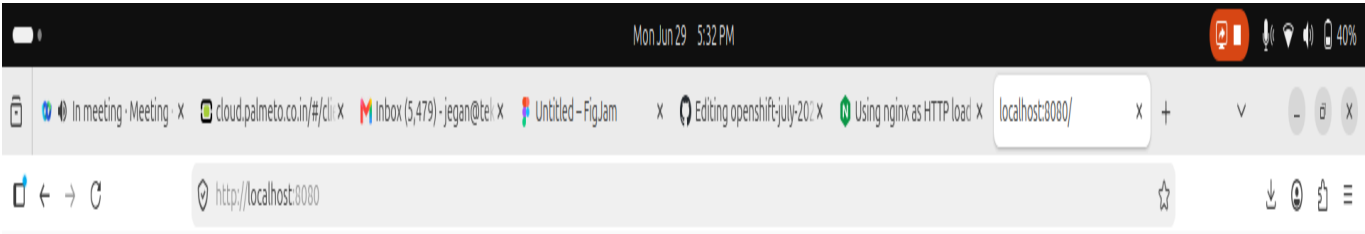
curl http://172.17.0.2:80
curl http://172.17.0.3:80
curl http://172.17.0.4:80
```

```
Mon Jun 29 5:27 PM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org:~$ echo "<h1>Web Server 1</h1>" > index.html
jegan@tektutor.org:~$ cat index.html
<h1>Web Server 1</h1>
jegan@tektutor.org:~$ docker cp index.html nginx1-jegan:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx1-jegan:/usr/share/nginx/html/index.html
jegan@tektutor.org:~$
jegan@tektutor.org:~$ echo "<h1>Web Server 2</h1>" > index.html
jegan@tektutor.org:~$ docker cp index.html nginx2-jegan:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx2-jegan:/usr/share/nginx/html/index.html
jegan@tektutor.org:~$
jegan@tektutor.org:~$ echo "<h1>Web Server 3</h1>" > index.html
jegan@tektutor.org:~$ docker cp index.html nginx3-jegan:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx3-jegan:/usr/share/nginx/html/index.html
jegan@tektutor.org:~$
jegan@tektutor.org:~$ curl http://172.17.0.2:80
<h1>Web Server 1</h1>
jegan@tektutor.org:~$
jegan@tektutor.org:~$ curl http://172.17.0.3:80
<h1>Web Server 2</h1>
jegan@tektutor.org:~$ curl http://172.17.0.4:80
<h1>Web Server 3</h1>
jegan@tektutor.org:~$
```

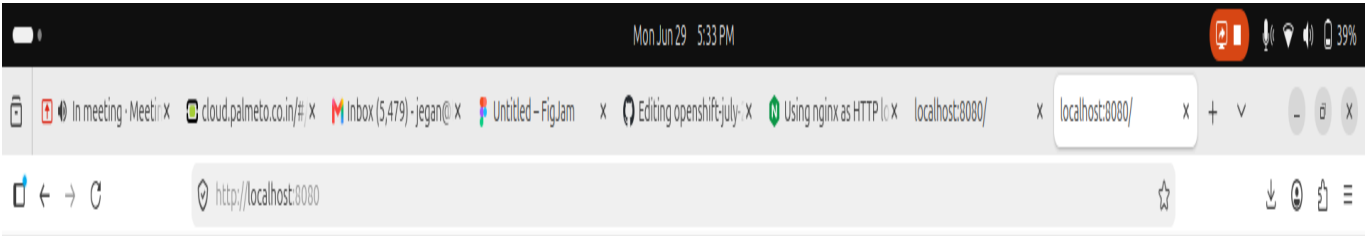


Let's verify if the lb-jegan container is working as configured(as a load balancer) from firefox web browser on your lab machine

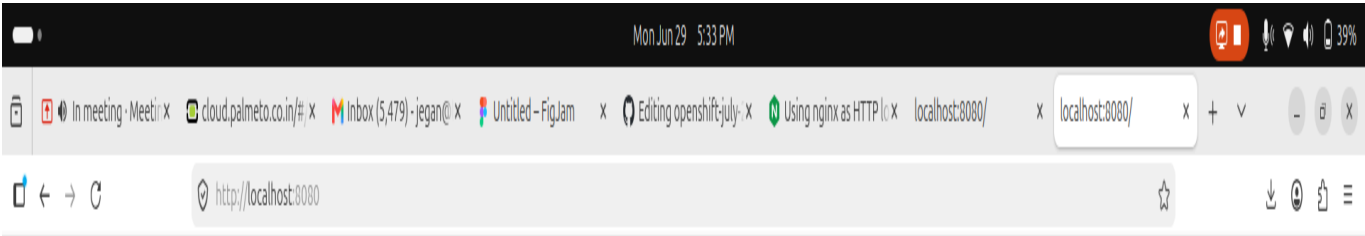
```
http://localhost:8080  
http://localhost:8080  
http://localhost:8080
```



Web Server 1



Web Server 2



Web Server 3

Checking the load balancer logs

```
docker logs lb-jegan
```

```
Mon Jun 29 5:35 PM
jegan@tektutor:~
jegan@tektutor:~
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
00b9192b12b8   nginx:latest   "/docker-entrypoint...." 23 minutes ago Up 5 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp lb-jegan
f43aac8a6630   nginx:latest   "/docker-entrypoint...." 31 minutes ago Up 31 minutes   80/tcp                               nginx3-jegan
86f7557ff0f3   nginx:latest   "/docker-entrypoint...." 31 minutes ago Up 31 minutes   80/tcp                               nginx2-jegan
8d1751568faa   nginx:latest   "/docker-entrypoint...." 31 minutes ago Up 31 minutes   80/tcp                               nginx1-jegan
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker logs lb-jegan
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/29 11:41:31 [notice] 1#1: using the "epoll" event method
2026/06/29 11:41:31 [notice] 1#1: nginx/1.31.1
2026/06/29 11:41:31 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/29 11:41:31 [notice] 1#1: OS: Linux 7.0.0-22-generic
2026/06/29 11:41:31 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/06/29 11:41:31 [notice] 1#1: start worker processes
2026/06/29 11:41:31 [notice] 1#1: start worker process 29
2026/06/29 11:41:31 [notice] 1#1: start worker process 30
2026/06/29 11:41:31 [notice] 1#1: start worker process 31
2026/06/29 11:41:31 [notice] 1#1: start worker process 32
2026/06/29 11:41:31 [notice] 1#1: start worker process 33
2026/06/29 11:41:31 [notice] 1#1: start worker process 34
2026/06/29 11:41:31 [notice] 1#1: start worker process 35
2026/06/29 11:41:31 [notice] 1#1: start worker process 36
```

```
2026/06/29 11:41:31 [notice] 1#1: start worker process 37
2026/06/29 11:41:31 [notice] 1#1: start worker process 38
2026/06/29 11:41:31 [notice] 1#1: start worker process 39
2026/06/29 11:41:31 [notice] 1#1: start worker process 40
2026/06/29 11:41:31 [notice] 1#1: start worker process 41
2026/06/29 11:41:31 [notice] 1#1: start worker process 42
```

Checking the webserver logs

```
docker logs nginx1-jegan
docker logs nginx2-jegan
docker logs nginx3-jegan
```

```

Mon Jun 29 5:36 PM
jegan@tektutor:~

jegan@tektutor:~
172.17.0.1 - - [29/Jun/2026:12:02:16 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:02:18 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:02:19 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:02:31 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:02:32 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:02:33 +0000] "GET / HTTP/1.1" 200 22 "-" "curl/8.18.0"
172.17.0.1 - - [29/Jun/2026:12:03:19 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:152.0) Gecko/20100101 Firefox/152.0"
172.17.0.1 - - [29/Jun/2026:12:03:22 +0000] "GET / HTTP/1.1" 200 22 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:152.0) Gecko/20100101 Firefox/152.0"
172.17.0.1 - - [29/Jun/2026:12:03:39 +0000] "GET / HTTP/1.1" 200 22 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:152.0) Gecko/20100101 Firefox/152.0"
jegan@tektutor.org:~$
jegan@tektutor.org:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
00b9192b12b8   nginx:latest   "/docker-entrypoint..." 24 minutes ago Up 6 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp lb-jegan
f43aac8a6630   nginx:latest   "/docker-entrypoint..." 32 minutes ago Up 32 minutes   80/tcp                               nginx3-jegan
86f7557ff0f3   nginx:latest   "/docker-entrypoint..." 32 minutes ago Up 32 minutes   80/tcp                               nginx2-jegan
8d1751568faa   nginx:latest   "/docker-entrypoint..." 32 minutes ago Up 32 minutes   80/tcp                               nginx1-jegan
jegan@tektutor.org:~$ docker logs nginx1-jegan
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/29 11:33:00 [notice] 1#1: using the "epoll" event method
2026/06/29 11:33:00 [notice] 1#1: nginx/1.31.1
2026/06/29 11:33:00 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)

```

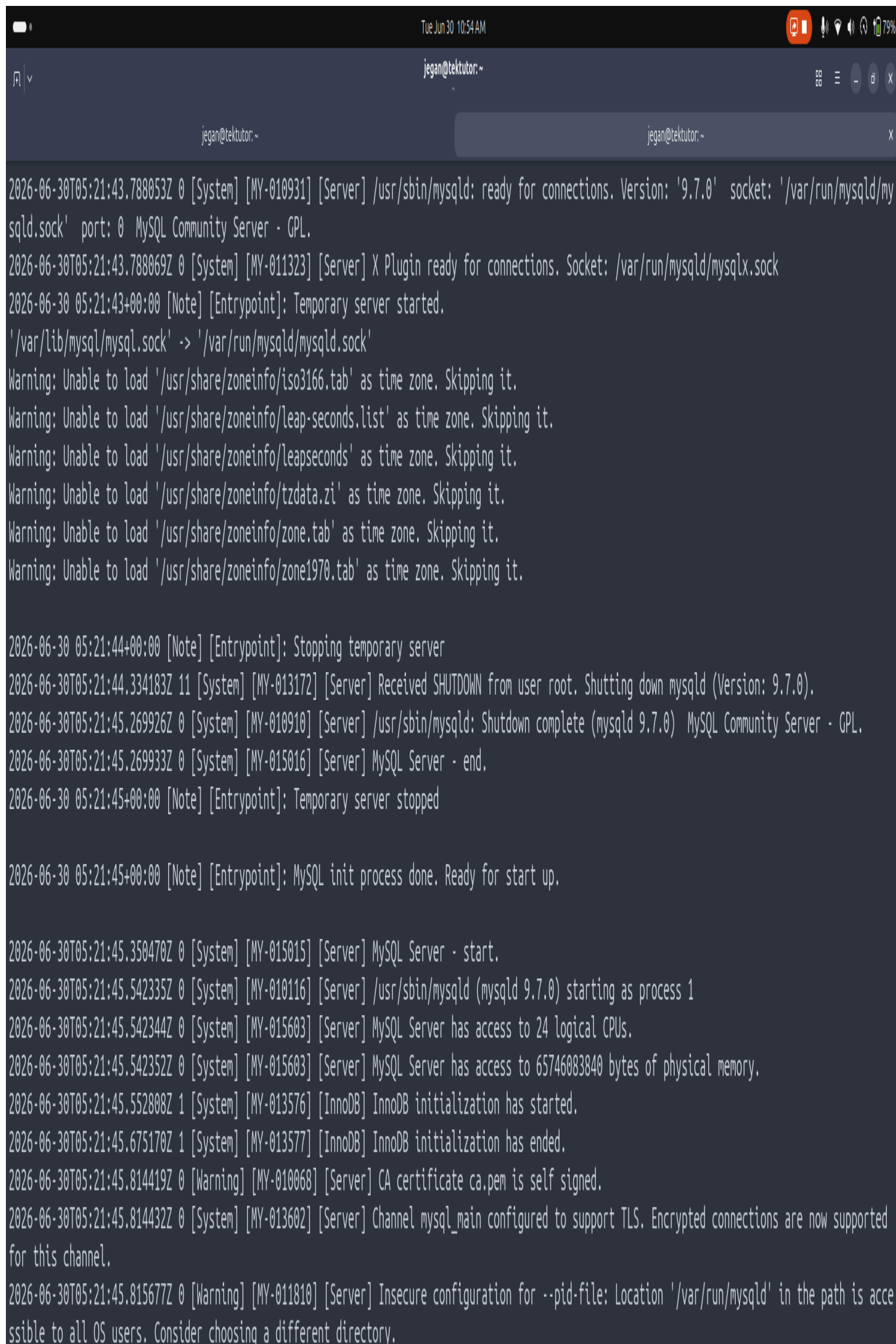
```
2026/06/29 11:33:00 [notice] 1#1: OS: Linux 7.0.0-22-generic
2026/06/29 11:33:00 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/06/29 11:33:00 [notice] 1#1: start worker processes
2026/06/29 11:33:00 [notice] 1#1: start worker process 29
2026/06/29 11:33:00 [notice] 1#1: start worker process 30
2026/06/29 11:33:00 [notice] 1#1: start worker process 31
```

Lab - Create a mysql db container

```
docker run -d --name mysql-jegan --hostname mysql-jegan -e
MYSQL_ROOT_PASSWORD=root@123 mysql:latest
docker ps
docker logs mysql-jegan
```

```
nt script for MySQL Server 9.7.0-1.el7
izing database files
ver] MySQL Server Initialization - st
ver] /usr/sbin/mysqld (mysqld 9.7.0)
oDB] InnoDB initialization has starte
oDB] InnoDB initialization has ended.
rver] root@localhost is created with

ver] MySQL Server Initialization - en
files initialized
temporary server
```

A terminal window with a dark background and light-colored text. The window title is 'jegan@tektutor:~'. The terminal shows the output of the MySQL service startup. It includes timestamps, log levels, and messages from the MySQL daemon and Entrypoint. The messages indicate that the MySQL server is ready for connections, a temporary server is started, various time zone files are skipped, the temporary server is stopped, and the main MySQL server starts successfully with access to 24 logical CPUs and 65746083840 bytes of physical memory. It also shows InnoDB initialization and a warning about a self-signed CA certificate.

```
Tue Jun 30 10:54 AM
jegan@tektutor:~

2026-06-30T05:21:43.788053Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.7.0' socket: '/var/run/mysqld/mysqld.sock' port: 0 MySQL Community Server - GPL.
2026-06-30T05:21:43.788069Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
2026-06-30 05:21:43+00:00 [Note] [Entrypoint]: Temporary server started.
'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.

2026-06-30 05:21:44+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-06-30T05:21:44.334183Z 11 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 9.7.0).
2026-06-30T05:21:45.269926Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 9.7.0) MySQL Community Server - GPL.
2026-06-30T05:21:45.269933Z 0 [System] [MY-015016] [Server] MySQL Server - end.
2026-06-30 05:21:45+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-06-30 05:21:45+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

2026-06-30T05:21:45.350470Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-06-30T05:21:45.542335Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 9.7.0) starting as process 1
2026-06-30T05:21:45.542344Z 0 [System] [MY-015603] [Server] MySQL Server has access to 24 logical CPUs.
2026-06-30T05:21:45.542352Z 0 [System] [MY-015603] [Server] MySQL Server has access to 65746083840 bytes of physical memory.
2026-06-30T05:21:45.552808Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-06-30T05:21:45.675170Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-06-30T05:21:45.814419Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-06-30T05:21:45.814432Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-06-30T05:21:45.815677Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
```



```
2026-06-30T05:21:45.827128Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.7.0' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
2026-06-30T05:21:45.827136Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
jegan@tektutor.org:~$
```

Let's get inside the mysql db server container

```
docker exec -it mysql-jegan /bin/sh
mysql -u root -p
SHOW DATABASES;
CREATE DATABASE tektutor;
USE tektutor;
CREATE TABLE trainings ( id INT NOT NULL, name VARCHAR(200) NOT NULL,
duration VARCHAR(200) NOT NULL, PRIMARY KEY(id) );
INSERT INTO trainings VALUES ( 1, "DevOps", "5 Days" );
INSERT INTO trainings VALUES ( 2, "Microservices with Golang", "5 Days" );
SELECT * FROM trainings;
exit
exit
docker ps
```

```
Tue Jun 30 10:57 AM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor.org: ~$ # Let's get inside the mysql db server container
jegan@tektutor.org: ~$ docker exec -it mysql-jegan /bin/sh
sh-5.1#
sh-5.1# hostname
sh: hostname: command not found
sh-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.0 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.002 sec)

mysql> CREATE DATABASE tektutor;
```

```
Query OK, 1 row affected (0.011 sec)
```

```
mysql> U
```

```
·>
```

```
·> ;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to u
```

```
Tue Jun 30 10:58 AM
jegan@tektutor: ~
jegan@tektutor: ~
->
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'U' at line 1
mysql> USE tektutor;
Database changed
mysql> SHOW t
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't' at line 1
mysql> SHOW TABLES;
Empty set (0.001 sec)

mysql> CREATE TABLE trainings ( id INT NOT NULL, name VARCHAR(200) NOT NULL, duration VARCHAR(200) NOT NULL, PRIMARY KEY(id) );
Query OK, 0 rows affected (0.018 sec)

mysql> INSERT INTO trainings VALUES ( 1, "DevOps", "5 Days" );
Query OK, 1 row affected (0.013 sec)

mysql> INSERT INTO trainings VALUES ( 2, "Microservices with Golang", "5 Days" );
Query OK, 1 row affected (0.011 sec)

mysql> SELECT * FROM trainings;
+----+-----+-----+
| id | name                | duration |
+----+-----+-----+
| 1  | DevOps              | 5 Days  |
| 2  | Microservices with Golang | 5 Days  |
+----+-----+-----+
2 rows in set (0.000 sec)
```

```
mysql> exit
Bye
sh-5.1# exit
exit
jegan@tektutor.org:~$
```

Let's try to restart the mysql container

```
docker restart mysql-jegan
docker ps
docker exec -it mysql-jegan /bin/sh
mysql -u root
SHOW DATABASES;
USE tektutor;
SHOW TABLES;
SELECT * FROM trainings;
exit
exit
```

```
jegan@tektutor:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
ab4009975448   mysql:latest   "docker-entrypoint.s..." 7 minutes ago  Up 7 minutes  3306/tcp, 33060/tcp      mysql-jegan

jegan@tektutor:~$ docker restart mysql-jegan
mysql-jegan

jegan@tektutor:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
ab4009975448   mysql:latest   "docker-entrypoint.s..." 9 minutes ago  Up 4 seconds  3306/tcp, 33060/tcp      mysql-jegan

jegan@tektutor:~$ docker exec -it mysql-jegan /bin/sh
sh-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.0 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| tektutor |
```

```
| tektutor |
+-----+
5 rows in set (0.009 sec)

mysql> USE tektutor;
Reading table information for completion of table and column names
```

```
Tue Jun 30 11:01 AM
jegan@tektutor: ~
jegan@tektutor: ~
jegan@tektutor: ~

+-----+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| tektutor            |
+-----+
5 rows in set (0.009 sec)

mysql> USE tektutor;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_tektutor |
+-----+
| trainings           |
+-----+
1 row in set (0.001 sec)

mysql> SELECT * FROM trainings;
+---+-----+-----+
| id | name                | duration |
+---+-----+-----+
| 1  | DevOps              | 5 Days  |
| 2  | Microservices with Golang | 5 Days  |
+---+-----+-----+
2 rows in set (0.000 sec)
```



```
mysql> exit
Bye
sh-5.1# exit
exit
jegan@tektutor.org:~$
```

Let's delete the mysql container. At this point, we not only lost the container, we also lost the data stored inside the container.

```
docker rm -f mysql-jegan
```

Hence, we must always store application data in an external storage.

```
mkdir -p /tmp/jegan/mysql
docker run -d --name mysql-jegan --hostname mysql-jegan -e
MYSQL_ROOT_PASSWORD=root@123 -v /tmp/jegan/mysql:/var/lib/mysql
mysql:latest
docker exec -it mysql-jegan /bin/sh
mysql -u root -p
SHOW DATABASES;
CREATE DATABASE tektutor;
USE tektutor;
CREATE TABLE trainings ( id INT NOT NULL, name VARCHAR(200) NOT NULL,
duration VARCHAR(200) NOT NULL, PRIMARY KEY(id) );
INSERT INTO trainings VALUES ( 1, "DevOps", "5 Days" );
INSERT INTO trainings VALUES ( 2, "Microservices with Golang", "5 Days" );
SELECT * FROM trainings;
exit
exit
docker ps
```

Let's delete the mysql-jegan container and recreate a new one

```
docker rm mysql-jegan
docker run -d --name mysql-jegan --hostname mysql-jegan -e
MYSQL_ROOT_PASSWORD=root@123 -v /tmp/jegan/mysql:/var/lib/mysql
mysql:latest
docker exec -it mysql-jegan /bin/sh
mysql -u root -p
SHOW DATABASES;
USE tektutor;
SHOW TABLES;
SELECT * FROM trainings;
```

```
exit  
exit
```

Though we delete the container and recreated a new one, we didn't lose the data because we stored the database in an external disk. This is how, containers are used in real-world applications.